

Experiment # 01

Construction of Straight-Through and Crossover Cable.

Objective

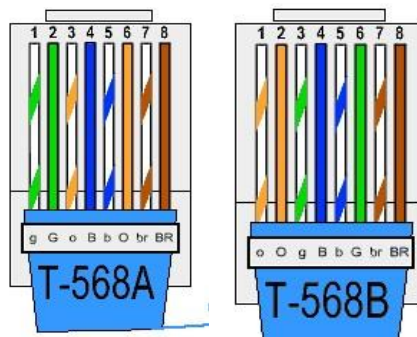
Differentiate and construct straight through and crossover cable.

Workbench/Workstation/Software Tools

- UTP cable
- Connector
- Crimper

Different types of devices are connected through straight-through cables. This type of cable is used more often. It can be used to connect a computer to a switch/hub's normal port. Straight cable has two sides, side A and side B. Each side has wire arrangement with same color. Crossover cable is used to connect the same type of devices. Rather, it can be used to connect two computers directly. Both types of this cable either straight cable or Cross cable has eight wires. There are four twisted pairs of these wires. Each wire has a different color code.

Cabling Standards



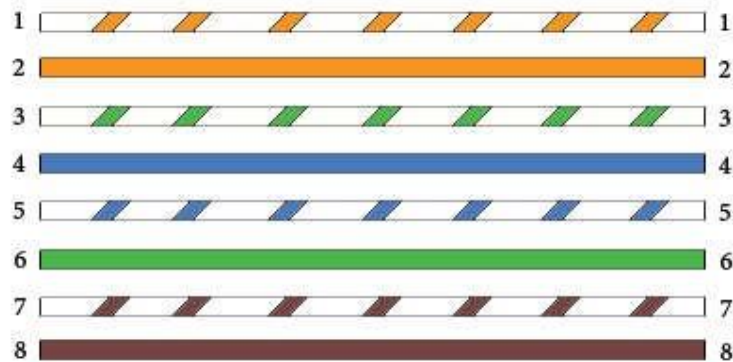
Cabling Types:

- Straight-through
- Crossover

Straight-Through Wired Cables

Straight-Through refers to cables that have the pin assignments on each end of the cable. In other words, Pin 1 connector A goes to Pin 1 on connector B, Pin 2 to Pin 2 etc. Straight-Through wired cables are most used to connect a host to client. When we talk about cat5e patch cables, the Straight-Through wired cat5e patch cable is used to connect computers, printers and other network client devices to the router switch or hub (the host device in this instance).

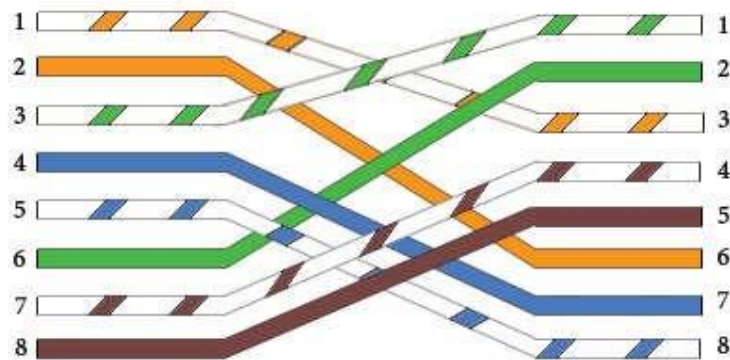
Straight Through Wiring Guide 568-B



Crossover Wired Cables

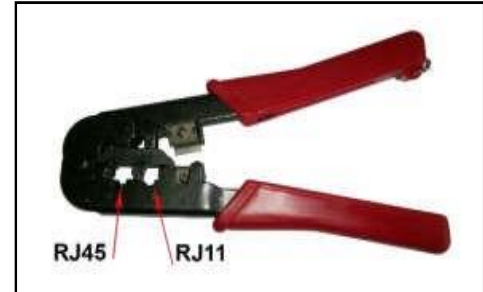
Crossover wired cables are very much like Straight-Through cables with the exception that TX and RX lines are crossed (they are at opposite positions on either end of the cable). Using the 568-B standard as an example below you will see that Pin 1 on connector A goes to Pin 3 on connector B. Pin 2 on connector A goes to Pin 6 on connector B etc. Crossover cables are mostly used to connect two hosts directly. Examples would be connecting a computer directly to another computer, connecting a switch directly to another switch, or connecting a router to a router. Note: While in the past when connecting two host devices directly a crossover cable was required. Now a days most devices have auto sensing technology that detects the cable and device and crosses pairs when needed.

Crossover Wiring Guide 568-B



Crimping Tool:

A crimping tool is a device used to conjoin two pieces of metal by deforming one or both in a way that causes them to hold each other. For instance, network cables and phone cables are created using a crimping tool (shown below) to join the RJ-45 and RJ-11 connectors to both ends of either phone or CAT5 cable.



Cable Tester:

Always remove the cable from both the computer and router or modem before connecting the tester. Most testers consist of two main components: the tester itself and the remote. Most testers have either two or three connectors: RJ-45 for Ethernet, RJ-11 for telephone cable, and BNC for coaxial cable. The smaller remote has two connectors, one each for RJ-11 and RJ-45. If the network cable tester supports coaxial cable networking, it comes with a terminator. Users simply plug the network cable into the appropriate port on the cable tester itself, and then connect the remote to the other end of the cable.



Reading a Network Cable Tester

The standard network cable tester has a row of 10 LEDs on the body and 9 LEDs on the remote to match up with the standard connector pinouts. The tester sends a signal down each pair in turn, lighting the LEDs up. The order in which they either light or do not light indicate the cable type and whether it is working.

Cable Condition	Reading
Straight Through	Light in same sequence on both units
Crossover	Light in reverse order on remote
Open Circuit	No lights on either main unit or remote
Short Circuit	Lights on main unit but not remote
4-Pair Telephone Wire	Only first four LEDs light and ground
6-Pair Telephone Wire	Only first six LEDs light and ground
Ethernet Cable	Eight LEDs plus ground all light in sequence
Coaxial Cable	Bottom LED and LED on terminator light